

CS23532 - COMPUTER NETWORKS LABORATORY

EXPERIMENT 3

Installation, Configuration, and Setup of Cisco Packet Tracer

1. AIM

To successfully download, install, configure, and set up Cisco Packet Tracer network simulation software on the host operating system, and to verify its operational workspace for designing, configuring, and troubleshooting network topologies.

2. PREREQUISITES

- Basic familiarity with computer operating systems (Windows / Linux / macOS).
- An active Cisco Networking Academy (SkillsForAll / NetAcad) user account for software authentication and access.
- Basic understanding of networking concepts (IP addresses, switches, routers, and host workstations).
- Administrative privileges on the target host computer to perform software installation.

3. APPARATUS / HARDWARE & SOFTWARE REQUIREMENTS

Component	Minimum Requirement	Recommended Specifications
Processor (CPU)	x86 64-bit compatible processor (1.5 GHz or higher)	Intel Core i5 / AMD Ryzen 5 or higher (2.0 GHz+)
System Memory (RAM)	4 GB RAM	8 GB RAM or higher
Storage Space	1.5 GB available disk space	3.0 GB available SSD storage space
Display Resolution	1024 x 768 resolution	1920 x 1080 (Full HD) resolution

Operating System	Microsoft Windows 10/11 (64-bit), Ubuntu Linux, or macOS	Windows 11 (64-bit) or latest Ubuntu LTS
Software Executable	Cisco Packet Tracer Standalone Installer (64-bit executable file)	Cisco Packet Tracer Standalone Installer (64-bit executable file)

4. STEP-BY-STEP PROCEDURE

Step 1: Downloading the Installer Package

1. Open a web browser and navigate to the official Cisco Networking Academy portal or SkillsForAll website.
2. Log in using your registered Cisco NetAcad / SkillsForAll credentials.
3. Navigate to the **Resources** tab and select **Download Packet Tracer**.
4. Select the appropriate installer executable for your host operating system (e.g., 64-bit Windows setup executable).
5. Save the installer file to your host computer's local storage.

[INSERT IMAGE HERE: Screenshot of Cisco NetAcad / SkillsForAll Download Page]

Figure 1: Cisco Networking Academy Download Page

Step 2: Executing the Software Installation Wizard

1. Locate the downloaded installer file (e.g., `CiscoPacketTracer822_64bit_setup.exe`) and right-click to select **Run as Administrator**.
2. When the License Agreement dialog appears, read the terms, select **I accept the agreement**, and click **Next**.
3. Specify the target installation directory (default path: `C:\Program Files\Cisco Packet Tracer`) and click **Next**.
4. Choose the Start Menu folder option and ensure the check box for **Create a desktop shortcut** is enabled. Click **Next**.
5. Review the configuration summary and click **Install** to initiate the extraction and installation process.
6. Once file extraction completes, check the option **Launch Cisco Packet Tracer** and click **Finish**.

[INSERT IMAGE HERE: Screenshot of Cisco Packet Tracer Setup Wizard / License Agreement Window]

Figure 2: Cisco Packet Tracer Installation Setup Wizard

Step 3: Initial Setup & Account Authentication

1. Upon initial launch, if prompted by Windows Defender Firewall, grant network access privileges for private networks.
2. When the Cisco Packet Tracer Login screen displays, enter your Cisco Networking Academy / SkillsForAll account credentials.
3. Check the **Keep me logged in** option to avoid re-authentication during future laboratory sessions.

[INSERT IMAGE HERE: Screenshot of Cisco Packet Tracer User Login Prompt]

Figure 3: Cisco Networking Academy User Authentication Screen

Step 4: Environment Verification & Basic Workplace Setup

1. Verify that the primary workspace components load correctly:
 - **Logical Workspace:** Central canvas for placing device icons.
 - **Bottom Device Toolbar:** Contains Routers, Switches, End Devices, Cables, and Security Appliances.
 - **Top Menu & Action Bar:** Selection, Inspection, Delete, and Note tools.
2. Perform a basic functionality test:
 - Drag two PC endpoints (**End Devices** → **PC**) onto the logical canvas.
 - Connect both PCs using a Copper Straight-Through or Auto-Connect Cable.
 - Click on each PC to assign static IP addresses (e.g., **192.168.1.1** and **192.168.1.2**).
 - Open the Command Prompt interface on PC0 and issue a **ping 192.168.1.2** command to confirm connectivity.

[INSERT IMAGE HERE: Screenshot of Packet Tracer Logical Workspace showing 2 connected PCs with successful Ping command output]

Figure 4: Simulated Topology and Successful Ping Verification in Cisco Packet Tracer

5. RESULT

Cisco Packet Tracer was successfully downloaded, installed, and configured on the host workstation. The application environment was authenticated via Cisco NetAcad credentials, and basic end-to-end network communication between host devices was successfully simulated and verified in the logical workspace.

